Project Description

October 2025

Overview

This document specifies a simplified model of strategic bidding in a liberalized electricity market and proposes a set of exercises that guide you from the bilevel formulation to implementable mixed-integer linear programs (MILPs) and computational experiments.

Deadline: 05/12/2025

Team up in groups of 2-3 people

Hand in: report in pdf and code, include a requirements.txt so all the packages you use are listed.

Problem Setting

Let $\mathcal{J}$ denote the set of strategic players and $\mathcal{I}$ the set of generators in the system. Player $j \in \mathcal{J}$ owns the subset of generators $\Omega_j^{\mathtt{I}} \subseteq \mathcal{I}$. Throughout the exercises you may assume $|\Omega_j^{\mathtt{I}}| = 1$ for clarity, i.e., each player owns one generator, although the notation supports multiple units per player.

Each generator $i \in \mathcal{I}$ can dispatch active power $P_i^{\mathtt{G}}$ within technical bounds

$$\mathrm{P}_i^{\mathtt{Gmin}} \le P_i^{\mathtt{G}} \le \mathrm{P}_i^{\mathtt{Gmax}} \qquad \forall i \in \mathcal{I}, \tag{1}$$

to collectively serve an inelastic system demand $\mathrm{D} > 0$. The unit marginal production cost is $\mathrm{c}_i^{\mathtt{G}}$.

Each generator submits a single price bid $\alpha_i^{\mathtt{G}}$ that must lie within market bounds

$$\alpha^{\mathtt{min}} \le \alpha_i^{\mathtt{G}} \le \alpha^{\mathtt{max}} \qquad \forall i \in \mathcal{I}. \tag{2}$$

Market clearing (lower level). Given bids $\alpha^{\mathtt{G}}$, the system operator clears the market by solving the economic dispatch:

$$\min_{X_{\mathrm{LL}}} \sum_{i\in\mathcal{I}} \alpha_i^{\mathtt{G}} P_i^{\mathtt{G}} \quad \text{s.t.} \quad \sum_{i\in\mathcal{I}} P_i^{\mathtt{G}} = \mathrm{D}, \quad \mathrm{P}_i^{\mathtt{Gmin}} \le P_i^{\mathtt{G}} \le \mathrm{P}_i^{\mathtt{Gmax}} \;\; \forall i \in \mathcal{I}, \tag{3}$$

with dual variables $\lambda_{n,t}$ (power balance) and $\mu_i^{\mathtt{G_{min}}}, \mu_i^{\mathtt{G_{max}}}$ (capacity bounds).

Strategic behavior (upper level). Each player $j \in \mathcal{J}$ chooses its bid(s) $\{\alpha_i^{\mathtt{G}}\}_{i\in\Omega_j^{\mathtt{I}}}$ to maximize profit, i.e., revenue at the cleared price minus production cost. For a single-generator player j this yields the canonical bilevel problem:

$$\min_{X_{\mathtt{UL}},\Xi_{\mathtt{UL}}} \left\{ \sum_{i\in\Omega_j^{\mathtt{I}}} \big(-\lambda P_i^{\mathtt{G}} + \mathrm{c}_i^{\mathtt{G}} P_i^{\mathtt{G}} \big) \right\} \tag{4a}$$

subject to:

$$\alpha^{\mathtt{min}} \le \alpha_i^{\mathtt{G}} \le \alpha^{\mathtt{max}} \quad \forall i \in \Omega_j^{\mathtt{I}}, \tag{4b}$$

$$\min_{X_{\mathrm{LL}}} \left\{ \sum_{i\in\mathcal{I}} \alpha_i^{\mathtt{G}} P_i^{\mathtt{G}} \right\} \tag{4c}$$

subject to:

$$\mathrm{D} - \sum_i P_i^{\mathtt{G}} = 0 : \lambda, \tag{4d}$$

$$\mathrm{P}_i^{\mathtt{Gmin}} \le P_i^{\mathtt{G}} \le \mathrm{P}_i^{\mathtt{Gmax}} : \mu_i^{\mathtt{G_{min}}}, \mu_i^{\mathtt{G_{max}}}, \qquad \forall i \in \mathcal{I}. \tag{4e}$$

Where:

$$X_{\mathtt{UL}} = \left\{ \alpha^{\mathtt{G}}_{i\in\Omega_j^{\mathtt{I}}}, P^{\mathtt{G}}_{i\in\mathcal{I}} \right\}$$
$$\Xi_{\mathtt{UL}} = \left\{ \lambda_{n,t} \mu_i^{\mathtt{G_{min}}}, \mu_i^{\mathtt{G_{max}}} \right\}$$
$$X_{\mathtt{LL}} = \left\{ P^{\mathtt{G}}_{i\in\mathcal{I}} \right\}$$

KKT reformulation. The lower-level problem in (4) is a linear program. It can be replaced by its Karush–Kuhn–Tucker (KKT) optimality conditions—stationarity, primal feasibility, dual feasibility, and complementary slackness—yielding a single-level mathematical program with complementarity constraints (MPCC). Complementarities can be linearized via either a big-M approach with binary variables or SOS1 constraints; the resulting model is a MILP once all bilinear terms are handled.

1 Exercise 1 (15 pt): Single-level MILP reformulation

a) Derive the KKT conditions of the lower-level dispatch and replace the lower-level problem in (4) with these conditions to obtain a single-level MPCC.

b) Linearize all complementary slackness relations using either (i) big-M with well-justified bounds, or (ii) SOS1 constraints. Explain your choice and the implications for tightness.

c) If your upper-level objective introduces bilinear terms, introduce auxiliary variables and apply linearizations, so that the final model is a MILP.

d) Organize your write-up in subsections: model derivation, complementarity handling, linearization details. Include a short *Nomenclature* listing all sets, parameters, variables, and dual variables used.

2 Exercise 2 (15 pt): Single-player experiments

Consider $\mathcal{I} = \{1, 2, 3, 4\}$ with the following guidelines:

- Choose capacities so that $\mathrm{P}_i^{\mathtt{Gmin}} = 0$ for all units and $\mathrm{P}_i^{\mathtt{Gmax}}$ values are small enough that three units are dispatched at the solution, out of the four. ~~,~~ ~~but avoid the knife-edge case D = $\sum_{i \in \mathcal{I}} P_i^{\mathtt{G}}$ at bounds.~~
- Take $\alpha^{\mathtt{min}} < 0$ and $\alpha^{\mathtt{max}} > 0$ broad enough not to be binding in typical cases.
- Let demand D be high enough so all units receive nonzero dispatch in merit order.

Solve (4) for a *single playing* player j in each of the scenarios below; other players' bids are fixed as indicated.

1. **Truthful competitors:** For all $i \notin \Omega_j^{\mathtt{I}}$, set $\alpha_i^{\mathtt{G}} = \mathrm{c}_i^{\mathtt{G}}$.
2. **Overbidding competitors:** For all $i \notin \Omega_j^{\mathtt{I}}$, set $\alpha_i^{\mathtt{G}} > \mathrm{c}_i^{\mathtt{G}}$ (justify your choice).

For each scenario, report:

a) A merit-order plot showing bids (prices), dispatch levels, and the demand intercept.

b) The playing player's profit.

3 Exercise 3 (15 pt): Best-Response (Diagonalization) among all players

Using the set-up of Exercise 2, compute a Nash equilibrium via iterative best responses ("Diagonalization Algorithm"). Initialize $\alpha_i^{\mathtt{G}} = \mathrm{c}_i^{\mathtt{G}}$ for all i at iteration $k = 0$. At iteration $k \geq 1$, update each player's bids by solving their bilevel problem while holding others' bids fixed at iteration $k - 1$.

Stopping criteria. Declare convergence when (i) no player can improve their own profit by more than 1% with a unilateral deviation, *and* (ii) bids stabilize within a user-chosen tolerance. Impose a maximum iteration cap suitable for your machine; if the cap is reached, stop and clearly label the outcome as "not converged".

Deliverables.

- Identify one configuration of $(\mathrm{c}_i^{\mathtt{G}}, \mathrm{P}_i^{\mathtt{Gmax}}, \mathrm{D}, \alpha^{\mathtt{min}}, \alpha^{\mathtt{max}})$ for which the algorithm converges.
- Plot, for each player, the sequence $k \mapsto \alpha_i^{\mathtt{G}}(k)$ together with the truthful cost $\mathrm{c}_i^{\mathtt{G}}$ and the final (converged) bid profile.
- Plot the dispatch of each generator at (i) the converged equilibrium and (ii) the centralized optimum obtained by clearing with true costs ("central dispatch"). Use bar charts.
- Compute the inefficiency of the equilibrium respect the central dispatch, report both numerator and denominator explicitly.

4 Exercise 4 (15 pt): Varying fuel costs

Using the convergent set-up identified in Exercise 3:

- For a system with exactly three players, compute the worst equilibrium inefficiency over a bounded *discrete* grid of $\mathrm{c}_i^{\mathtt{G}}$ values per generator. State the grid and bounds.
- Repeat the analysis for systems with $|\mathcal{I}| \in \{4, \dots, 10\}$ players/generators. Clearly define how capacities and demand scale with $|\mathcal{I}|$.

Visualize at least:

a) Equilibrium inefficiency as a function of the number of players $|\mathcal{I}|$.

b) The share (in %) of instances that converged before hitting the iteration cap.

5 Exercise 5 (40 pt): Open-ended investigation

Choose **one** topic and perform a thorough, well-argued analysis (model choices, parameterization, diagnostics, and discussion):

- Impact of market price bounds $(\alpha^{\mathtt{min}}, \alpha^{\mathtt{max}})$ on the inefficiency of the equilibrium that can be obtained via the Best Response algorithm.
- Sensitivity of the inefficiency of the equilibrium respect to demand level D (e.g., low, medium, peak).
- Effect of asymmetric capacities or costs across players on convergence and inefficiency of the equilibrium.
- Comparison of big-M versus SOS1 linearizations in terms of tightness and runtime.

- Policy-equilibrium concept. How does it differ from the bids-equilibrium, and how policies perform.
- Any other topic you find relevant. Propose and motivate it. Needs to be approved before it can be developed.

Nomenclature (suggested)

Sets: $\mathcal{J}$ (players), $\mathcal{I}$ (generators), $\Omega_j^{\mathtt{I}} \subseteq \mathcal{I}$ (units owned by j).
Parameters: D (demand), $\mathrm{c}_i^{\mathtt{G}}$ (marginal cost), $\mathrm{P}_i^{\mathtt{Gmin}}, \mathrm{P}_i^{\mathtt{Gmax}}$ (capacity bounds), $\alpha^{\mathtt{min}}, \alpha^{\mathtt{max}}$ (market price bounds).
Decision variables: $P_i^{\mathtt{G}}$ (dispatch), $\alpha_i^{\mathtt{G}}$ (bid).
Dual variables: $\lambda_{n,t}$ (balance), $\mu_i^{\mathtt{G_{min}}}, \mu_i^{\mathtt{G_{max}}}$ (capacity bounds).

Implementation tips. Use consistent units; document all bounds used in big-M formulations; seed random experiments for reproducibility; and report solver, version.